8-30

Z+T Ch 3 S 1

Classification of first-order PDE's by linearity.

ODE: sol'n is a f'n of one indep. variable

PDE: sol'n is a f'n of two or more indep. variables

indep. variable x or t

almost always dots ⇒ time is indep. variable

examples $y'' + e^x y' + y = 3$ $\ddot{x}\, x = e^{t}$

dep: z

indep. variables x, y

$z_{xx} + z_{xy} - x^2 z = e^x \sin(y)$

$u e^v w_{uu} - v^2 w_v = \sin(v)$

dep. variable $w_{uu} = \frac{\partial^2 w}{\partial u^2}$

indep: u, v

⇒ sol'n is a surface.

$z = x^2 + y^2$

<u>Def'n</u> A <u>first-order PDE</u> in 2 indep. variables, (x,y), and one dep. variable, z, is an eq'n of the form

$$(*) \quad F(x, y, z, z_x, z_y) = 0$$

$F(\quad) = \frac{1}{2} z_x + \frac{z}{z_y} xy = 5$

ex) $x e^y z^{z_x} - e^{z_y} \sin(x + z_y) = 0$

<u>Def'n</u> A sol'n to $(*)$ is a <u>function</u>

surface $\rightarrow$ $z = f(x,y) \in C^1(\Omega)$ such that

(i) for every $(x,y) \in \Omega$, (x, y, z, z_x, z_y) is in the domain of F

(ii) when $z = f(x,y)$ is subst. into $(*)$, an identity results.

Classification

(i) <u>Quasi-linear</u> coeff of z_x, z_y, can be a f'n of z

$$P(x,y,z)\,z_x + Q(x,y,z)\,z_y = R(x,y,z)$$

(ii) <u>Almost linear</u> Linear z_x, z_y

$$P(x,y)\,z_x + Q(x,y)\,z_y = \underbrace{R(x,y,z)}_{\text{nonlinear f'n in } z}$$

(iii) <u>Linear</u> Linear in z_x, z_y, z

$$\underbrace{a(x,y)}_{\text{indep. variables only}}\,z_x + \underbrace{b(x,y)}\,z_y + \underbrace{c(x,y)}\,z = \underbrace{d(x,y)}_{\text{indep. variables only}}$$

(iv) <u>Nonlinear</u> Not (i), (ii), or (iii)

Classify

ex 1) $z\frac{\partial z}{\partial x} + e^x \frac{\partial z}{\partial y} = xy$

dep. variable: z
indep: x, y

Quasi linear

$P(x,y,z) = z$
$Q(x,y,z) = e^x$
$R(x,y,z) = xy$

ex 2) $y z_x + e^x z_y = xyz$

$y z_x + e^x z_y - xyz = 0$

Linear

$a(x,y) = y \quad b(x,y) = e^x \quad c(x,y) = -xy$

ex 3) $y\frac{\partial u}{\partial x} + e^x \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \underbrace{xz^2}_{\text{only a fn of indep. variables}}$

dep. variable u
indep. variable x, y, z

Linear

ex 4) $y\frac{\partial z}{\partial x} + e^x \frac{\partial z}{\partial y} = \underbrace{xyz^2}_{R}$

dep. variable z
indep. variables: x, y

$P = y \quad Q = e^x$

Almost Linear

ex 5) $e^x \frac{\partial z}{\partial x}\frac{\partial z}{\partial y} = xyz^2$

Nonlinear eq'n

Method of characteristics

A method for solving 1st-order quasilinear PDE's given an initial curve.

Note: 1st-order ODE, need initial pt. (IC)
1st-order PDE, " " curve (IC)

Consider

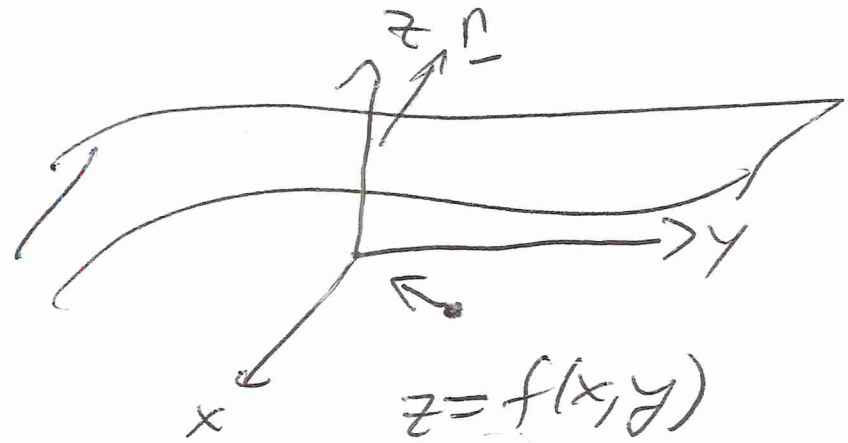
(*) $P(x,y,z)\, z_x + Q(x,y,z)\, z_y = R(x,y,z)$

The sol'n $\underbrace{z = f(x,y)}$ is a surface in $\mathbb{R}^3$

could $F(x,y,z) = 0$
will need IFT, $\frac{\partial F}{\partial z} \neq 0$

Recall

i) Given a surface, $F(x,y,z)=0$,
The vector $\underline{\nabla} F$ is normal to surface everywhere.



$z=f(x,y)$

$F(x,y,z)=z-f(x,y)=0$

(ii) A parametric curve
$\langle x(\tau), y(\tau), z(\tau)\rangle$
has a tangent line
$\langle x'(\tau), y'(\tau), z'(\tau)\rangle$

So consider a soln $z=z(x,y)$
It can also be written as a level surface
$$F(x,y,z) = \overset{f(x,y)}{z(x,y)} - z = 0$$
Normal to this level surface $= \underline{\nabla} F = \langle z_x, z_y, -1\rangle$

① By geometry, any curve $\langle x(s), y(s), z(s) \rangle$ that has one pt. on the sol'n surface, and whose tangent, $\langle x'(s), y'(s), z'(s) \rangle$ is everywhere $\perp$ to $\langle z_x, z_y, -1 \rangle$ remains on the surface $F(x, y, z) = 0$.